

# National University of Computer and Emerging Sciences



## Lab Exercise 07 AL2002-Artificial Intelligence Lab

|                   |                                                                                      |  |
|-------------------|--------------------------------------------------------------------------------------|--|
| Lab Instructor(s) |  |  |
| Semester          | Spring 2026                                                                          |  |

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## Exercise

A thief enters a shop carrying bag which can carry 35 kgs of weight. The shop has 10 items, each with a specific weight and price. Now, the thief's dilemma is to make such a selection of items that it maximizes the value (i.e., total price) without exceeding the knapsack weight. We have to help the thief to make the selection.

Available Items are:

| Item No. | Weight | Value |
|----------|--------|-------|
| 1        | 3      | 266   |
| 2        | 13     | 442   |
| 3        | 10     | 671   |
| 4        | 9      | 526   |
| 5        | 7      | 388   |
| 6        | 1      | 245   |
| 7        | 8      | 210   |
| 8        | 8      | 145   |
| 9        | 2      | 126   |
| 10       | 9      | 322   |

Initial population:

```
[[0 1 0 1 1 0 0 1 1 1]
 [1 1 1 1 0 1 1 1 0 0]

 [0 1 0 0 0 0 1 1 0 1]

 [0 0 1 0 1 1 0 0 0 0]

 [0 0 1 1 0 0 0 0 0 1]
```

[0 1 0 1 1 0 1 0 0 0]

[1 1 1 0 0 0 1 0 1 0]

[0 0 0 0 1 1 1 0 0 0]

Fitness Function:

$$fitness = \sum_{i=1}^n c_i v_i; \text{ if } \sum_{i=1}^n c_i w_i \leq kw$$

$$fitness = 0; \text{ otherwise}$$

where,

n = chromosome length Ci = ith gene

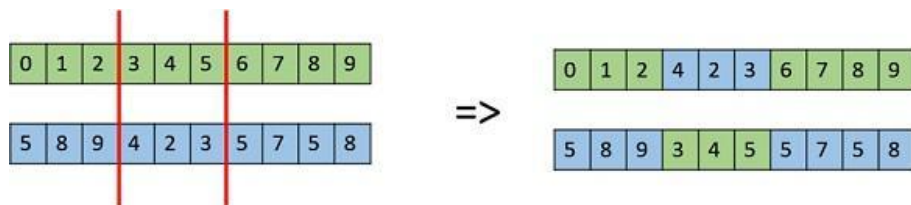
Vi = ith value Wi = ith weight

kw = allowed weight

To generate new population:

1. Two-point crossover

Example of two-point cross-over. Choose the positions as in the example.



2. Mutate offspring got from cross over at 2 random positions. Select 2 random positions: if you find 1 change it to 0 and vice versa.

Always keep 8 best chromosomes, 50% from initial population and 50% from new generation. Fixed the number of iterations to 15 or 20. Output the best chromosome at the end.

**Task: Use genetic algorithm to help the thief select appropriate items.**